

Physical AI operating leader · machine intelligence, quality, and adoption

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<https://russelldudek.github.io/Xora/>

VP OF ARTIFICIAL
INTELLIGENCE · 4MP

Immediate contribution at the machine- intelligence boundary.

Operator-technologist who makes complex physical work visible, connects technical and operating teams, and builds mechanisms that turn emerging technology into adopted, measurable outcomes.

**Make every correction
explainable, bounded,
validated, and reusable.**

4MP needs an AI leader who can connect manufacturing intent, machine state, metrology evidence, physics constraints, human authority, and reusable learning.

Physical systems

Robotics, machine vision, telemetry, field feedback

AI in the work

Agents, retrieval, evaluation, escalation, governance

Quality under consequence

Root cause, process control, standard work

Founding mechanisms

Architecture, prototypes, roadmaps, team cadence

SELECTED EXPERIENCE

2025–PRESENT

Vape-Jet

Director of Operations

- Leads AI-first operating transformation across production, purchasing, quality, customer success, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry, and AI-assisted workflows.
- Connects factory, field, customer, machine-fleet, quality, support, and engineering signals into prioritized work, earlier risk detection, cleaner handoffs, dashboards, standard work, training, and accountability.
- Works directly in a software-enabled robotics and machine-vision environment with approximately 1,000 fielded robots informing support and operating feedback loops.

2017–PRESENT

DudeWorth

Founder / AI and Automation
Advisor

- Develops AI-readiness and opportunity frameworks across value, feasibility, data readiness, governance, adoption burden, reuse, and time-to-value.
- Designs agentic workflow patterns using structured context, retrieval, AI agents, human judgment, escalation, decision records, evaluation, and governance rails.
- Advises on AI-enabled operations, robotics, logistics technology, warehouse automation, AMRs, ASRS, cobots, and transportation systems.

2000–2013

Compunetics

Director of Operations and
Engineering Management

- Led engineering, manufacturing, quality, customer delivery, and complex technical programs in advanced electronics and high-reliability environments.
- Built quality systems, standard work, root-cause practices, performance measures, and process controls.
- Collaborated with high-consequence customers and partners including DoD, CERN, Intel, and AMD.

Machine + telemetry

Physical-system signals into operating action

Authority + evidence

Human review, evaluation, and decision records

Manufacturing + quality

Root cause and process control

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COMPLEMENTARY DEPTH

Autonomy, scale, scientific grounding, and technical range.

AUTONOMY AND OPERATING SCALE

2018–2022

ZeusVu · Chief Pilot, Autonomous Systems and Aerial Data Operations

Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents. Developed TensorFlow-based concepts for automated aerial inventory and computer-vision field intelligence.

2014–2018

Amazon · Operations Management / Site Leader

Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward environment supporting approximately five million second-day orders annually. Built leadership routines, labor planning, throughput visibility, escalation, standard work, and Lean/Gemba mechanisms.

2021–2023

Cardinal Building Products · Regional Director

Built the company's first remote regional operation and modernized distributed customer, CRM, fulfillment, logistics, digital marketing, SEO, and online-buying workflows.

SCIENTIFIC AND TECHNICAL FOUNDATION

University of Pittsburgh · Bachelor of Science

Academic grounding includes materials science, physical chemistry, and biology-related studies. Academic work is indexed through Russell's Google Scholar profile.

Technical communication and ecosystem leadership

Published industry articles, delivered technical talks, participated in the HyperTransport Consortium, and led regional IEEE programming connecting industry, academia, electronics, manufacturing, and emerging technology.

CREDENTIALS

Google AI Essentials

Food Safety Management Certification

IBM Enterprise Design Thinking Practitioner

Six Sigma Certification

OSHA 10

TECHNICAL AND OPERATING TOOLKIT

Agentic AI

LLM workflows

RAG patterns

Human-in-loop

Python

SQL

JavaScript

TensorFlow

Tableau

ERP/Odoo

Jira

Confluence

HubSpot

Slack

GIS

WHAT THIS RECORD SUPPORTS AT 4MP

Translate across AI, software, physical systems, manufacturing, quality, and customer reality.

Connect observation, root cause, human authority, corrective action, evidence, and reusable operating knowledge.

Architect mechanisms, build prototypes, review technical choices, recruit capability, and make adoption part of system design.

Day-one value: integration authority across machine, data, quality, and work.

The candidate vision includes the Qualification Lens, a 120-day plan, and a Correction Qualification Record.